

GOSTOMEL, Ukraine

WMO: 333451

Lat: 50.603N

Lon: 30.192E

Elev: 158

StdP: 99.45

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-15.0	-22.0	0.5	-17.1	-18.2	0.8	-14.1	9.9	2.4	9.0	2.1	2.6	40	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.1	19.8	30.2	19.3	28.8	18.7	21.7	27.6	20.9	27.0	20.0	25.9	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.0	23.8	19.0	14.1	23.0	18.1	13.2	22.2	64.1	28.0	61.1	27.3	58.3	25.5	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.8	7.0	-21.6	34.8	4.1	2.1	-24.5	36.3	-27.0	37.5	-29.3	38.7	-32.3	40.2		
			-21.8	23.0	4.1	1.4	-24.7	24.0	-27.2	24.8	-29.5	25.6	-32.5	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-4.0	-3.2	2.3	10.0	15.9	19.5	21.0	20.3	15.0	8.5	3.5	-0.9
	DBStd	9.98	6.01	6.04	4.85	4.06	4.16	3.34	2.97	3.67	3.93	4.25	4.32	4.87
	HDD10.0	1723	434	370	242	49	5	0	0	0	5	79	200	339
	HDD18.3	3661	692	603	498	253	98	27	9	21	113	304	446	597
	CDD10.0	1376	0	0	2	48	188	283	342	319	156	34	4	0
	CDD18.3	272	0	0	0	2	22	61	93	81	13	0	0	0
	CDH23.3	2752	0	0	0	22	247	576	857	894	154	1	0	0
	CDH26.7	901	0	0	0	5	54	161	284	362	34	0	0	0
Wind	WSAvg	3.4	3.8	3.5	4.0	3.5	3.2	3.2	3.0	2.9	3.0	3.2	3.6	3.9
Precipitation	PrecAvg	594	35	35	36	40	60	77	73	58	53	43	44	41
	PrecMax	747	56	75	105	113	159	158	157	117	178	110	90	133
	PrecMin	442	9	9	5	2	20	11	26	6	0	0	2	9
	PrecStd	71	13	16	23	24	33	37	33	31	46	27	21	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	10.1	17.1	26.8	30.2	32.1	34.0	36.0	30.1	22.9	16.1	10.9
		MCWB	6.2	7.0	10.3	15.6	18.3	19.6	21.0	19.5	18.2	14.6	12.4	9.4
	2%	DB	6.1	6.8	14.1	22.1	28.1	30.1	31.8	33.1	27.0	20.0	13.2	8.2
		MCWB	5.0	4.6	8.1	12.7	16.8	18.8	20.1	19.9	16.7	14.4	11.1	7.2
	5%	DB	4.8	4.9	11.8	19.8	26.1	28.8	29.9	30.8	25.0	17.2	11.1	6.9
		MCWB	3.8	3.3	6.7	11.9	16.3	18.6	20.1	19.0	16.3	12.7	9.2	6.0
	10%	DB	3.0	3.1	9.2	17.8	24.0	26.9	28.0	28.1	22.8	15.0	9.8	5.1
		MCWB	2.4	1.9	5.6	10.8	16.0	18.2	19.1	18.2	15.5	11.9	8.5	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	7.9	10.4	15.8	20.3	22.0	23.3	22.6	19.7	16.2	13.0	9.8
		MCDB	7.4	9.8	16.1	24.9	27.7	26.9	29.5	29.1	27.9	20.1	15.1	10.3
	2%	WB	5.1	5.0	8.9	14.0	18.6	20.8	22.1	21.3	18.2	14.8	11.6	7.5
		MCDB	5.6	6.2	13.3	20.2	24.6	26.6	27.9	27.9	24.6	18.5	13.4	8.3
	5%	WB	3.6	3.4	7.3	12.7	17.6	19.9	21.2	20.3	17.1	13.5	9.8	5.8
		MCDB	4.2	4.4	10.1	17.7	23.5	25.6	26.9	26.9	22.8	16.6	10.9	6.4
	10%	WB	2.2	2.1	5.9	11.5	16.5	19.1	20.3	19.5	16.1	12.1	8.4	4.4
		MCDB	2.8	2.8	8.8	16.5	22.2	24.9	25.6	26.0	21.1	14.9	9.4	5.0
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.7	7.7	10.3	11.1	11.1	11.1	11.6	10.6	8.6	4.9	4.3
		MCDBR	4.8	6.7	11.9	13.5	13.9	13.6	13.6	15.1	14.2	12.0	7.4	6.0
		MCWBR	4.3	5.0	7.0	6.5	5.5	5.1	5.0	4.8	6.1	6.9	5.6	4.9
	5% WB	MCDBR	4.6	5.8	10.1	11.5	12.1	11.6	11.3	12.3	12.2	10.6	6.4	5.3
		MCWBR	4.2	4.5	6.7	6.0	5.5	5.4	4.8	5.0	5.9	6.7	5.3	4.6
Clear-Sky Solar Irradiance	taub	0.307	0.339	0.371	0.433	0.408	0.400	0.429	0.432	0.399	0.360	0.341	0.308	
	taud	2.213	2.158	2.213	2.130	2.284	2.327	2.257	2.237	2.288	2.373	2.308	2.252	
	Ebn at Noon	698	770	820	806	849	858	824	799	783	750	657	638	
	Edh at Noon	73	101	115	140	126	121	128	125	107	82	68	61	
All-Sky Solar Radiation	RadAvg	0.85	1.67	2.96	4.32	5.62	5.95	5.74	5.00	3.42	1.96	0.78	0.57	
	RadStd	0.15	0.30	0.47	0.44	0.46	0.44	0.39	0.31	0.51	0.32	0.11	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.75	N/A	N/A	+0.97	+0.58	+0.54	-140	-202	+117	+52

Nomenclature: See separate page